

A DERIVATION IN SEVEN PARTS

Gradient Residuals

The Calculus of Clearance

Research Note VI: The Geometry of Volatility

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Gradientology

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CORE VOLATILITY CONSTANTS (DERIVED IN THIS ESSAY)

$$\begin{array}{lcl} R = \sigma/c = 40 & \cdot & \text{BRF exhaust} = 4/3 \quad \cdot \quad I(\text{exhaust}) = \log_2(3) \text{ bit/step} \quad \cdot \\ & & N_{\text{sat}} = n^2/c_{\text{steps}} = 25 \end{array}$$

Abstract

Essay VI derives the geometric structure of nothing's kinetic volatility — the quantitative account of how much of nothing's structural space remains permanently inaccessible to the helical worldline, why it remains inaccessible, and what structural consequences this inaccessibility forces. The three clearances I_{min} , σ , Λ , established as static spatial margins in Essays I–IV and inverted into kinetic functions in Essay V, are now read as kinetic volatilities: the exact fractions of nothing's structural space that the helical worldline structurally cannot fill at each ordinal position. Their geometric behaviour under the self-application of the Inverted Ratio generates three new structural results that could not be derived from the static or kinematic registers alone.

Part 0 establishes the three volatilities as kinetic complements: $V_S = I_{min} = 1/5$ (Source amplitude deficit), $V_B = \sigma = 2/5$ (Boundary denominator deficit), $V_R = \Lambda = 0.2984$ (Remainder drive deficit). Part I derives the Minimum Volatility Theorem (T.MVT): the Boundary volatility σ must strictly exceed the Source volatility I_{min} — $\sigma > I_{min}$ — for the kinetic margin Φ to remain positive. This is a purely algebraic necessity from nothing's clearances, with no free parameters. Part II derives the Boundary Resolution Frame ($R = \sigma/c = 40$) as the volatility macro-period: the ordinal span in which the helical worldline traverses exactly one Boundary volatility width axially. Part III derives the exact arc coverage: per ordinal step, the worldline attempts to cover $G_{raw}/\delta = 28/3 = 9.333$ lattice positions of arc; it addresses 9 (G_{snap}/δ); the permanent sub-lattice gap is $1/3$ per step. Part IV derives the Volatility Identity: $\Lambda = (1 - d_R) - \Phi$ — the Remainder volatility equals the Remainder arc deficit minus the kinetic margin. This is a new structural identity not previously established. Part V derives the Forbidden Ceiling as the exact kinetic exhaust rate: per ordinal step, the structural exhaust $\varepsilon_{snap} = 1/30$ carries information content $-\log_2(\varepsilon_{snap}/\delta) = \log_2(3) =$ the Forbidden Ceiling. The kinetic structure exhausts exactly the Forbidden information per step — the informational expression of why nothing's self-application can never saturate its own forbidden ceiling. Part VI derives the Saturation Threshold $N_{sat} = n^2/c_{steps} = 25$ from nothing's lattice combinatorics.

Keywords

Kinetic Volatility · Minimum Volatility Theorem · Boundary Resolution Frame · Arc Coverage · Sub-Lattice Gap · Volatility Identity · Forbidden Ceiling Exhaust Rate · Saturation Threshold · Volatility Hierarchy · Exhaust Accumulation

Introduction

From Clearance to Volatility

In the static registers of Essays I–III, the three clearances $I_{min} = 1/5$, $\sigma = 2/5$, and $\Lambda = 0.2984$ were structural margins: they were the portions of each axis that nothing retained as clearance against the structural failure modes (Generative Seizure, Deterministic Crystal, Causal Closure Seizure). They were defined by what they prevented — the failure modes at their boundaries — and by their precise lattice positions within nothing's structural space $NS = [0,1]^3$.

Essay V inverted the three clearances into kinetic functions. I_{min} became the cosine amplitude deficit. σ 's complement d_B became the regulatory divisor generating the Chronon and causality speed. Λ became the permanent drive deficit keeping δ below Ω . The clearances were not abolished by the inversion — they were transformed. Their values were preserved; their structural roles changed.

Essay VI asks the question that the inversion raises but does not yet answer: what does the kinetic structure DO with the clearances? How much of nothing's structural space does the helical worldline cover, how much does it permanently leave uncovered, and what forces the uncovered regions to remain uncovered? The answer is the geometry of volatility: the exact quantitative account of the permanent inaccessibility of each clearance fraction in the kinetic register, and the three new structural theorems that this account generates.

What Is Groundbreaking in Essay VI

Three results in Essay VI are new — not re-readings of Essay I–V results but genuinely new derivations that could not be made without the full kinetic structure of Essay V.

The Minimum Volatility Theorem (T.MVT): The algebraic condition for positive kinetic margin $\Phi > 0$ is $\sigma > I_{min}$. The Boundary volatility must strictly exceed the Source volatility. This is a cross-clearance constraint: one clearance is bounded below by another. It was invisible in the static register (where each clearance was constrained by its own failure modes, not by the other clearances) and becomes visible only in the kinetic register where both appear as volatility fractions in the Inverted Ratio's output equation.

The Volatility Identity (T.VI): $\Lambda = (1 - d_R) - \Phi$. The Remainder volatility equals the Remainder amplitude deficit (the fraction of the Remainder axis not swept by the helix) minus the kinetic margin. This identity connects the static geometry ($1 - d_R = 3/10$, the fraction of the Remainder axis above the helix's maximum amplitude), the kinetic dynamics ($\Phi = 0.0016$), and the static clearance ($\Lambda = 0.2984$) in a single algebraic equation.

The Forbidden Ceiling as Exhaust Rate (T.FCE): Per ordinal step, the sub-lattice exhaust $\varepsilon_{snap} = 1/30$ carries information content exactly equal to $\log_2(3)$ — the Forbidden Ceiling of Essay III. This means: nothing's kinetic structure exhausts exactly the forbidden information per step. The kinetic exhaust IS the information that T.NE forbids the static structure from achieving. The static information bound and the kinetic exhaust rate are structurally identical.

Part 0 — The Three Kinetic Volatilities

Part 0 defines the three kinetic volatilities as the kinetic expressions of the three clearances, and establishes the conceptual framework within which the rest of Essay VI operates.

0.1 Volatility as the Kinetic Complement

In the static register, a clearance is the fraction of an axis that nothing retains as margin — the axis territory that is not occupied by the face's co-primary density. In the kinetic register, this same fraction is a volatility: the fraction of the structural range that the helical worldline permanently does not occupy at the axis's extremes. The two descriptions — clearance in the static register, volatility in the kinetic register — refer to the same structural fraction, but the structural meaning differs. A clearance is a spatial margin preventing collapse. A volatility is a geometric gap ensuring the worldline cannot saturate the axis.

The key distinction: in the static register, the clearance is the absence of the co-primary density. In the kinetic register, the volatility is the deficit between the worldline's maximum reach on that axis and the field ceiling $\Omega = 1$. These are the same numbers ($I_{min} = 1/5$, $\sigma = 2/5$, $\Lambda = 0.2984$) because the co-primary density on each axis equals the worldline's maximum reach on that axis — but they are structurally different descriptions. The static clearance prevents the structural failure modes. The kinetic volatility prevents the trajectory failure modes (rotational saturation, axial seizure, causal saturation).

Definition D.KV · The Three Kinetic Volatilities

The three kinetic volatilities, each derived from the corresponding clearance:

$$\begin{aligned} V_S &= I_{min} = 1/5 = 0.2000 \\ &\text{(Source volatility: cosine amplitude deficit)} \end{aligned} \tag{1}$$

The fraction by which the Source component max = $d_S = 4/5$ falls short of $\Omega = 1$

$$\begin{aligned} V_B &= \sigma = 2/5 = 0.4000 \\ &\text{(Boundary volatility: denominator deficit)} \end{aligned} \tag{2}$$

The fraction by which the Boundary co-primary density $d_B = 3/5$ falls short of $\Omega = 1$

$$V_R = \Lambda = 0.2984$$

(Remainder volatility: drive deficit) (3)

The fraction by which the Order Parameter $\delta = 0.7016$ falls short of $\Omega = 1$

Each volatility is the same number as the corresponding clearance (I_{min} , σ , Λ) because $d_S = 1 - I_{min}$, $d_B = 1 - \sigma$, and $\delta = 1 - \Lambda$ by co-primacy (T.CP, Essay I).

0.2 The Volatility Hierarchy

The clearance hierarchy established in Essay I (P.HI) and confirmed in Essays II and III — $\sigma > \Lambda > I_{min}$ — is the same as the volatility hierarchy. Since the volatilities ARE the clearances, the hierarchy transfers directly:

Principle P.VH · The Volatility Hierarchy

From P.HI (Essay I) and T.FE (Essay III):

$$V_B > V_R > V_S \quad (4)$$

$$\sigma > \Lambda > I_{min} \quad (5)$$

$$2/5 > 0.2984 > 1/5 \quad (6)$$

$$0.4000 > 0.2984 > 0.2000 \quad (7)$$

The Boundary face retains the largest volatility (the most structural space permanently inaccessible to the worldline on the Boundary axis), consistent with its role as the regulatory divisor in the Inverted Ratio — the largest clearance governs the most constraint on the self-application. The Source face retains the smallest volatility, consistent with its role as the phase-leading generative component — it has the smallest deficit from Ω . The Remainder is intermediate.

0.3 The Trajectory Failure Modes as Volatility Boundaries

Each volatility has two failure boundaries — one at $V=0$ and one at $V=1$ — corresponding to the two structural failure modes for that face. These were derived as trajectory failure modes in Essay V (T.SCI for Source, T.BDN dual foreclosure for Boundary, T.PKR for Remainder). Essay VI now gives them their volatility expression:

Principle P.VFM · Volatility Failure Modes

Source volatility V_S :

$$V_S = 0 \quad (d_S=1): \text{ cosine amplitude} = \Omega \Rightarrow \text{rotational saturation at each peak} \quad (8)$$

$$V_S \rightarrow 1 \quad (d_S \rightarrow 0): \text{ cosine amplitude} = 0 \Rightarrow \text{no Source component} \Rightarrow \text{Triad collapses} \quad (9)$$

Boundary volatility V_B :

$$V_B = 0 \quad (d_B=1): G_{\text{raw}} = d_S * d_R \Rightarrow \Phi < 0 \Rightarrow \text{kinetic seizure (T.BCL)} \quad (10)$$

$$V_B \rightarrow 1 \quad (d_B \rightarrow 0): G_{\text{raw}} \rightarrow \text{infinity} \Rightarrow \text{no lattice expression} \Rightarrow \text{dissolution (T.BCU)} \quad (11)$$

Remainder volatility V_R :

$$V_R = 0 \quad (\delta=1): \text{ Order Parameter saturates } \Omega \Rightarrow \text{Causal Closure Seizure (T.RCL)} \quad (12)$$

$$V_R \rightarrow 1 \quad (\delta \rightarrow 0): TI^\beta \rightarrow 0 \Rightarrow \text{no kinetic output} \Rightarrow \text{stasis (T.SCU)} \quad (13)$$

All six failure boundaries are foreclosed by the same structural constraints (T.TS, T.TD, T.RCL, T.SCU) that foreclosed them in the static register. The kinetic volatility boundary is the same as the static clearance boundary.

Part I — The Minimum Volatility Theorem — σ Must Exceed I_{min}

Part I derives the Minimum Volatility Theorem: a cross-clearance constraint that was invisible in the static register but is structurally necessary in the kinetic register. The theorem states that the Boundary volatility must strictly exceed the Source volatility for the kinetic margin to remain positive. This is a genuinely new structural result — it connects the two largest clearances through an algebraic condition that neither could reveal individually.

I.1 The Algebraic Condition for Positive Kinetic Margin

The kinetic margin $\Phi = \delta - d_R = TI^\beta - d_R$ must be strictly positive for the kinetic output to be nonzero. $\delta = TI^\beta = (d_S \times d_R \times d_B)^\beta$. The kinetic margin positivity condition is:

Derivation V.MVD · The Kinetic Margin Positivity Condition

$\Phi > 0$ requires $\delta > d_R$:

$$TI^\beta > d_R \quad (1)$$

$$(d_S \times d_R \times d_B)^\beta > d_R \quad (2)$$

For $\beta = 13/40 < 1$, the power law $(d_S \times d_R \times d_B)^\beta < (d_S \times d_R \times d_B)$ for any value in $(0,1)$. The condition becomes:

$$d_S \times d_R \times d_B \text{ vs } d_R \\ (\text{both being positive values in } (0,1)) \quad (3)$$

For $\Phi > 0$ to hold in the power-law form, an equivalent sufficient condition from the actual values is obtained by examining what controls the sign of $\delta - d_R$. $\delta = TI^\beta$ increases with TI . $TI = d_S \times d_R \times d_B$. As d_B decreases ($V_B = \sigma$ increases), TI decreases, δ decreases — which drives Φ negative. As d_B increases ($V_B = \sigma$ decreases), TI increases, δ increases — which helps Φ become positive.

The G_{raw} analysis (Essay V) showed that at $\sigma = 0$, $d_B = 1$, $G_{raw} = 0.56 < d_R = 0.7$ — a negative $G_{raw} - d_R$. But $\Phi = \delta - d_R$, not $G_{raw} - d_R$. δ is derived from TI via the power law, not directly from G_{raw} .

$$G_{raw} - d_R = 0.56 - 0.7 = -0.14$$

(negative: G_{raw} falls below d_R)

(8)

$$\delta - d_R = 0.8094 - 0.7 = +0.1094$$

(positive: δ above d_R even at $\sigma=0$)

(9)

The seizure at $\sigma = 0$ is correctly identified as $G_{raw} < d_R$ (the ratio itself drops below the threshold), not as $\delta < d_R$. The kinetic seizure is a ratio-level collapse, not a power-law collapse. The correct seizure condition is: $G_{raw} < d_R$, which occurs when σ falls below the critical threshold derived next.

I.2 The Correct Seizure Condition: G_{raw} vs d_R

The correct kinetic seizure condition is $G_{raw} < d_R$ — the Inverted Ratio falls below the structural threshold. This is the condition that makes the 'drive' of the Inverted Ratio insufficient to exceed the Remainder threshold. The Inverted Ratio $G_{raw} = (d_S \times d_R)/d_B$. The threshold is d_R . Setting $G_{raw} = d_R$ as the critical boundary:

Derivation V.GRC · The G_{raw} Seizure Threshold

$G_{raw} = d_R$ (critical boundary):

$$(d_S \times d_R) / d_B = d_R$$
(10)

$$d_S / d_B = 1$$
(11)

$$d_S = d_B$$
(12)

$G_{raw} > d_R$ requires $d_S > d_B$. Since $d_S = 1 - I_{min}$ and $d_B = 1 - \sigma$:

$$1 - I_{min} > 1 - \sigma$$
(13)

$$\sigma > I_{min} \quad (14)$$

The Inverted Ratio exceeds the structural threshold if and only if the Boundary volatility exceeds the Source volatility.

I.3 T.MVT – The Minimum Volatility Theorem

Theorem T.MVT · The Minimum Volatility Theorem

The kinetic structure of the Inverted Ratio $G = (d_S \times d_R)/d_B$ maintains structural viability ($G_{raw} > d_R$, the Inverted Ratio exceeds the threshold) if and only if:

$$\sigma > I_{min} \quad (15)$$

$$V_B > V_S \quad (16)$$

$$\text{Boundary volatility} > \text{Source volatility} \quad (17)$$

This is the Minimum Volatility Theorem. Its derivation:

$$G_{raw} > d_R \quad (18)$$

$$(d_S \times d_R) / d_B > d_R \quad (19)$$

$$d_S > d_B \quad (\text{dividing both sides by } d_R > 0) \quad (20)$$

$$1 - I_{min} > 1 - \sigma$$

(substituting $d_S = 1 - I_{min}$, $d_B = 1 - \sigma$) (21)

$$\sigma > I_{min} \quad (\text{rearranging}) \quad (22)$$

Verification: $\sigma = 2/5 = 0.4 > I_{min} = 1/5 = 0.2$. ✓ The Boundary volatility exceeds the Source volatility. The Inverted Ratio is viable.

Structural significance: T.MVT is a cross-clearance constraint invisible in the static register. In the static register, σ and I_{min} were each constrained by their own failure modes independently.

T.MVT reveals that in the kinetic register, the two clearances are jointly constrained: if σ were reduced to equal I_{min} , the Inverted Ratio would collapse to $G_{raw} = d_R$, producing $\Phi = 0$ (zero output). The kinetic structure requires the Boundary volatility to exceed the Source volatility by a strictly positive margin — and the actual value $\sigma/I_{min} = (2/5)/(1/5) = 2$ (forced by $c/a = 2$ from T.UC, Essay I) provides exactly this margin.

I.4 The G_{raw} Margin as a Volatility Ratio

The Inverted Ratio $G_{raw} = (d_S \times d_R)/d_B$ can be rewritten entirely in terms of the three volatilities:

Derivation V.GVM • G_{raw} Expressed in Terms of Volatilities

Since $d_S = 1 - I_{min} = 1 - V_S$, $d_B = 1 - \sigma = 1 - V_B$, and $d_R = 1 - (1 - d_R)$ with $(1 - d_R) = 0.3 \approx V_R + \Phi$:

$$G_{raw} = (1 - V_S) \times d_R / (1 - V_B) \quad (23)$$

At the critical boundary $\sigma = I_{min}$ ($G_{raw} = d_R$):

$$G_{raw|critical} = (1 - I_{min}) \times d_R / (1 - I_{min}) = d_R \quad (24)$$

The excess above d_R :

$$\begin{aligned} G_{raw} - d_R &= d_S \times d_R / d_B - d_R = \\ d_R \times (d_S/d_B - 1) &= d_R \times (d_S - d_B)/d_B \end{aligned} \quad (25)$$

Since $d_S - d_B = (1 - I_{min}) - (1 - \sigma) = \sigma - I_{min}$:

$$G_{raw} - d_R = d_R \times (\sigma - I_{min}) / d_B \quad (26)$$

$$= (7/10) \times (2/5 - 1/5) / (3/5) \quad (27)$$

$$= (7/10) \times (1/5) / (3/5) \quad (28)$$

$$= (7/10) \times (1/3) \quad (29)$$

$$= 7/30 = 0.2333 \quad (30)$$

The excess $G_{raw} - d_R = 7/30 = 0.2333$. Note: this is the RAW ratio excess, not the kinetic margin $\Phi = \delta - d_R = 0.0016$. The power law compresses $7/30$ to 0.0016 via the exponent $\beta = 13/40 < 1$. The raw excess is structurally substantial ($7/30 \approx 23.3\%$ of d_R); the power-law compression reduces it to the tight margin $\Phi = 0.0016$.

I.5 Foreclosure of $\sigma = I_{min}$

Theorem T.MVF · Foreclosure: $\sigma = I_{min}$

Suppose $\sigma = I_{min} = 1/5$. Then $d_B = 1 - \sigma = 4/5 = d_S$. From T.NE (Essay I): $d_S \neq d_B$. $d_S = d_B$ is foreclosed by T.NE. Therefore $\sigma = I_{min}$ is structurally foreclosed by T.NE — independently of T.MVT.

$$\begin{aligned} \sigma = I_{min} &\Rightarrow d_B = d_S \Rightarrow \text{T.NE violated} \Rightarrow \\ \text{Triad destroyed} & \end{aligned} \quad (31)$$

T.MVT provides the kinetic foreclosure (zero output) and T.NE provides the ontological foreclosure (equal face measures). Both independently foreclose $\sigma = I_{min}$. The actual value $\sigma = 2/5 > I_{min} = 1/5$ satisfies both constraints with margin $\sigma - I_{min} = 1/5 > 0$.

Part II — The Boundary Resolution Frame as Volatility Period

Part II derives the Boundary Resolution Frame ($R = 40$) as the volatility macro-period — the natural ordinal span over which the Boundary volatility's geometric consequences fully manifest. The result $R = \sigma/c = 40$ is an exact identity between the Boundary volatility, the causality speed, and the Boundary Resolution Frame derived in Essay I. This is not a coincidence: it is the algebraic expression of the fact that $c = \delta/\tau_0$ and $R = \sigma/\delta$ (in step units), giving $R = \sigma * \tau_0 / \delta = \sigma / c$.

II.1 The Boundary Resolution Frame as σ/c

Theorem T.BRV · The Boundary Resolution Frame as Volatility-Speed Ratio

From Essay I (V.BR): $R = n \times c_steps = 10 \times 4 = 40$.

From Essay V (V.CS): $c = \delta/\tau_0 = \delta^2 = 0.01$.

From the volatility definition: $\sigma = c_steps \times \delta = 4 \times 0.1 = 0.4$.

Therefore:

$$R = \sigma / c = (2/5) / (1/100) = (2/5) \times 100 = 40 \quad (1)$$

Verification: $R = n \times c_steps = 10 \times 4 = 40 \checkmark$. The two derivations of R agree.

Structural reading: R is the number of ordinal steps required for the helical worldline to advance axially by exactly $\sigma = V_B$. One Boundary Resolution Frame is the ordinal span in which the worldline traverses one Boundary volatility width axially.

II.2 What the Worldline Does in One *BRF*

In one *BRF* of $R = 40$ ordinal steps, the helical worldline:

Derivation V.BRFD · Worldline Activity in One Boundary Resolution Frame

Rotational:

$$R / \tau_0 = 40 / 10 = 4$$

complete rotations of the elliptic helix (2)

Axial:

$$R \times c = 40 \times 0.01 = 0.4 = \sigma$$

(one Boundary volatility width) (3)

Arc coverage (G_{snap} -addressable positions):

$$R \times G_{snap} = 40 \times (9/10) = 36$$

G_{snap} -addressed positions per BRF (4)

Sub-lattice exhaust accumulated:

$$R \times eps_{snap} = 40 \times (1/30) = 40/30 = 4/3$$

(5)

$$\text{As a fraction of } \delta: (4/3) / (1/10) = 40/3 = 13.33$$

δ -steps of exhaust per *BRF* (6)

Information exhaust per *BRF*:

$$R \times \log_2(3) = 40 \times 1.585 = 63.40 \text{ bits per BRF}$$

(7)

$$= \log_2(3^{40})$$

(information content of 40 forbidden-ceiling steps) (8)

The *BRF* is precisely the ordinal span that balances the Boundary volatility geometrically: the worldline enters the Boundary clearance region (z in $[3/5, 1]$) after exactly $3/5 / c = 60$ ordinal steps = 1.5 *BRFs*, and exits it ($z = 1 = \Omega$, axial closure) after 10 *BRFs*. One *BRF* spans one-tenth of the full axial closure (100 ordinal steps).

II.3 The *BRF* as Nothing's Natural Structural Period

The Boundary Resolution Frame $R = 40$ emerges from the derivation chain at three distinct levels:

Principle P.BRN · The *BRF* at Three Structural Levels

Level 1 — Lattice arithmetic (Essay I, V.BR): $R = n \times c_steps = 10 \times 4 = 40$.

Level 2 — Kinetic geometry (Essay V, Part II): $R = \tau_0 \times (R/\tau_0) = \tau_0 \times \sigma_steps = 10 \times 4 = 40$.

Level 3 — Volatility dynamics (Essay VI, T.BRV): $R = \sigma / c = (2/5) / (1/100) = 40$.

All three derivations give $R = 40$. The *BRF* is the natural structural period of nothing's kinetic volatility — the span at which the Boundary volatility is completely traversed axially. It is not a chosen period; it is the unique ordinal span at which the Boundary volatility σ and the causality speed c reconcile into an integer: $R = \sigma/c = (2/5)/(1/100) = 40$.

Part III — The Exact Arc Coverage and the Sub-Lattice Gap

Part III derives the exact arc coverage of the helical worldline at every ordinal step, the permanent sub-lattice gap between the attempted coverage and the addressable coverage, and the accumulation period of this gap. The arc coverage analysis is the core geometric content of the Geometry of Volatility: it reveals precisely where the worldline covers nothing's structural space and where it permanently does not.

III.1 Arc Coverage Definitions

The 'arc coverage' of the helical worldline at ordinal step τ is the number of distinct lattice positions that the worldline has visited or traversed between ordinal step 0 and ordinal step τ . On the discrete $n = 10$ lattice, lattice positions are indexed by integer multiples of $\delta = 1/10$. The worldline's position at step τ is $\gamma(\tau) = (d_S \cos(\Omega^*\tau), d_R \sin(\Omega^*\tau), c^*\tau)$. Coverage is measured in terms of the Inverted Ratio G_{raw} , since $G_{raw} = (d_S \times d_R)/d_B$ is the ratio that governs how much of the structural space the worldline 'activates' per step in the axial-rotational projection.

Per ordinal step, the worldline's coverage potential is G_{raw}/δ lattice positions: this is how many lattice positions of the $n = 10$ scale are 'reached' by the Inverted Ratio in one step. Since $G_{raw} = 14/15$ and $\delta = 1/10$, the coverage potential per step is $G_{raw}/\delta = 140/15 = 28/3 = 9.333$ positions. The addressable coverage is $\text{floor}(28/3) = 9$ positions (G_{snap}/δ). The sub-lattice gap is $1/3$ position per step.

Definition D.ARC · Arc Coverage Definitions

Per ordinal step τ :

Coverage potential:

$$G_{raw} / \delta = (14/15) / (1/10) = 28/3 = 9.3333 \text{ positions} \quad (1)$$

Addressable coverage:

$$G_{snap} / \delta = (9/10) / (1/10) = 9 \text{ positions} \quad (2)$$

Sub-lattice gap:

$$\text{eps_snap} / \delta = (1/30) / (1/10) = 1/3 \text{ position} \quad (3)$$

Per Chronon ($\tau_0 = 10$ steps):

Coverage potential:

$$10 \times 28/3 = 280/3 = 93.3333 \text{ positions} \quad (4)$$

$$\text{Addressable coverage: } 10 \times 9 = 90 \text{ positions} \quad (5)$$

$$\begin{aligned} \text{Sub-lattice gap per Chronon: } 10 \times (1/3) &= 10/3 = \\ 3.3333 \text{ positions} \end{aligned} \quad (6)$$

The sub-lattice gap accumulates across ordinal steps. After k steps: $k/3$ positions of gap have accumulated. After $k = 3$ steps: 1 full position of gap (minimum integer accumulation). After $k = \tau_0 = 10$ steps: $10/3$ positions, containing 3 integer positions and $1/3$ residual.

III.2 The Permanent Sub-Lattice Gap

The sub-lattice gap of $1/3$ position per ordinal step is permanent and irremovable. It arises because $G_{\text{raw}}/\delta = 28/3$ is not an integer — the Inverted Ratio, divided by the lattice grain, cannot be expressed as a whole number of lattice steps. This non-integrality is a structural consequence of the Boundary inversion: $d_B = 3/5$ appears in the denominator of G_{raw} , and 3 is the count of the Triad's faces ($d = 3$ from T.3D, Essay II) — a number that divides neither $n = 10$ nor $n^2 = 100$ without remainder.

Theorem T.PSG · The Permanent Sub-Lattice Gap

$G_{\text{raw}} / \delta = 28/3$ is non-integer because:

$$\begin{aligned} G_{\text{raw}} &= d_S \times d_R / d_B = (4/5) (7/10) / (3/5) = \\ (14/25) / (3/5) &= 14/15 \end{aligned} \quad (7)$$

$$G_{raw} / \delta = (14/15) / (1/10) = 140/15 = 28/3 \quad (8)$$

$28/3$ is non-integer because 3 (the dimension count $d = 3$ from T.3D) does not divide $28 = 4 \times 7 = (2^2)(7)$. Specifically: $\gcd(28, 3) = 1$ — 28 and 3 are coprime. Therefore $28/3$ is an irreducible fraction, and $1/3$ cannot be expressed as an integer number of δ -steps. The sub-lattice gap is $1/3$ per step — permanently non-integer.

The structural source: $d_B = 3/5$ appears in the denominator. $d_B = 1 - \sigma = 1 - 2/5 = 3/5 = c_{steps_complement} / n$. The 3 in the denominator of $G_{raw}/\delta = 28/3$ is the Triadic dimension count $d = 3$. The permanent sub-lattice gap is the geometric consequence of nothing's three-dimensionality ($d = 3$) being incommensurate with the $n = 10$ lattice.

III.3 The Gap Accumulation Period

The sub-lattice gap accumulates across ordinal steps at a rate of $1/3$ per step. The minimum ordinal span at which the accumulated gap reaches an integer number of lattice positions is the fundamental closure period of the gap:

Derivation V.GAP · The Gap Accumulation Period

After k ordinal steps, accumulated gap = $k/3$ positions.

For this to be an integer: k must be a multiple of 3. Smallest: $k = 3$.

$$\begin{aligned} k = 3: \quad & \text{accumulated gap} = 1 \text{ lattice position} \\ & (\text{minimum integer closure}) \end{aligned} \quad (9)$$

$$\begin{aligned} k = 3n: \quad & \text{accumulated gap} = n \text{ lattice positions} \\ & (\text{general closure}) \end{aligned} \quad (10)$$

The gap closure condition $k \times (1/3) \equiv 0 \pmod{1}$ is equivalent to the closure condition $k \times \text{eps}_{snap} \equiv 0 \pmod{\delta}$ derived in Essay V for the fundamental stable knot: $k \times (1/30) \equiv 0 \pmod{1/10}$. These are the same condition.

Gap closure at $k = 3$ is the geometric expression of the Fundamental Stable Knot $k_{min} = 3$ (Essay V, D.KCR). The minimum ordinal span for the arc coverage gap to close is exactly $k_{min} = 3$ — the number that characterises the minimum stable recursive structure.

III.4 The Cumulative Arc Coverage Table

The cumulative arc coverage (addressable positions) and sub-lattice deficit (positions permanently unaddressed) after key ordinal milestones:

Derivation V.CAC · Cumulative Arc Coverage at Key Milestones

After $\tau = 1$ (1 step):

$$\text{Addressable: } 9 \quad | \quad \text{Gap: } 1/3 \quad | \quad \text{Cumulative gap: } 1/3 \quad (11)$$

After $\tau = 3$ (3 steps, first gap closure):

$$\begin{array}{l} \text{Addressable: } 27 \quad | \quad \text{Gap: } 1 \quad (\text{first full integer gap step}) \\ \quad \quad \quad | \quad \text{Cumulative: } 1 \end{array} \quad (12)$$

After $\tau = 10 = \tau_0$ (one Chronon):

$$\begin{array}{l} \text{Addressable: } 90 \quad | \quad \text{Gap: } 10/3 = 3.33 \quad | \\ \quad \quad \quad \text{Full gap steps: } 3 \quad | \quad \text{Residual: } 1/3 \end{array} \quad (13)$$

After $\tau = 30 = 3 \times \tau_0$ (three Chronons, n-closure):

$$\begin{array}{l} \text{Addressable: } 270 \quad | \quad \text{Cumulative gap: } 10 = n \\ \quad \quad \quad (\text{gap has accumulated } n \text{ full positions}) \end{array} \quad (14)$$

After $\tau = 40 = R$ (one BRF):

$$\begin{array}{l} \text{Addressable: } 360 \quad | \quad \text{Cumulative gap: } 40/3 = 13.33 \\ \quad \quad \quad | \quad \text{Full gap steps: } 13 \quad | \quad \text{Residual: } 1/3 \end{array} \quad (15)$$

After $\tau = 100 = \tau_0^2$ (full axial closure):

$$\begin{array}{lcl} \text{Addressable: } 900 & | & \text{Cumulative gap: } 100/3 = 33.33 \\ & | & \text{Full gap steps: } 33 \quad | \quad \text{Residual: } 1/3 \end{array} \quad (16)$$

The residual $1/3$ persists even after 100 ordinal steps. It is never resolved. The permanent sub-lattice gap of $1/3$ is irreducible.

Part IV — The Volatility Identity — Λ Precisely Located

Part IV derives the Volatility Identity: a new algebraic relationship connecting the three volatilities through the kinetic margin. The identity $\Lambda = (1 - d_R) - \Phi$ states that the Remainder volatility equals the Remainder amplitude deficit (the gap between the helix's maximum Remainder-axis reach and the field ceiling) minus the kinetic margin. This is a structural decomposition of Λ into two geometrically distinct contributions.

IV.1 The Remainder Amplitude Deficit

The helical worldline has Remainder-axis amplitude $d_R = 7/10$. The maximum reach of the Remainder component $y(\tau) = d_R \sin(\Omega \tau)$ is $d_R = 0.7$. The deficit between this maximum reach and the field ceiling $\Omega = 1$ is:

Derivation V.RAD · The Remainder Amplitude Deficit

From Essay V (D.HWL):

$$\begin{aligned} y(\tau)_{\max} &= d_R = 7/10 = 0.700 \\ &\text{(maximum Remainder-component of worldline)} \end{aligned} \tag{1}$$

$$\begin{aligned} \text{Remainder amplitude deficit} &= \\ \Omega - d_R &= 1 - 7/10 = 3/10 = 0.300 \end{aligned} \tag{2}$$

This $3/10$ is the fraction of the Remainder axis above $d_R = 7/10$ — the region the helix cannot reach on the Remainder axis. Comparing to the Remainder volatility $V_R = \Lambda = 0.2984$:

$$1 - d_R = 0.3000 \tag{3}$$

$$\Lambda = 0.2984 \tag{4}$$

$$\begin{aligned} \text{Difference: } (1 - d_R) - \Lambda &= 0.0016 = \Phi \\ &\text{(the kinetic margin)} \end{aligned} \tag{5}$$

IV.2 T.VI — The Volatility Identity

Theorem T.VI • The Volatility Identity

The Remainder volatility Λ equals the Remainder amplitude deficit minus the kinetic margin:

$$\Lambda = (1 - d_R) - \Phi \quad (6)$$

$$0.2984 = 0.3000 - 0.0016 \quad \text{ok} \quad (7)$$

Derivation:

$$\Lambda = 1 - \delta \quad (\text{definition, Essay V T.PKR}) \quad (8)$$

$$\begin{aligned} &= 1 - (d_R + \Phi) \\ &\quad (\text{since } \delta = d_R + \Phi \text{ by definition of } \Phi) \end{aligned} \quad (9)$$

$$= (1 - d_R) - \Phi \quad (10)$$

Interpretation: the Remainder volatility has two components. The first $(1 - d_R = 3/10)$ is the amplitude deficit: the fraction of the Remainder axis that the helix's Remainder component cannot reach geometrically because its amplitude is $d_R = 7/10 < 1$. The second $(-\Phi = -0.0016)$ is the kinetic margin adjustment: the Order Parameter $\delta = 0.7016$ exceeds $d_R = 0.7000$ by exactly Φ , which means δ enters slightly into the amplitude deficit region. $\Lambda = (1 - d_R) - \Phi$ is the amplitude deficit reduced by the kinetic margin — the amount by which the drive exceeds the threshold is subtracted from the geometric deficit.

Structural significance: T.VI reveals that Λ is not an independent structural quantity — it is algebraically determined by d_R and Φ , both of which are derived from d_S, d_R, d_B , and β . Λ 's value is the geometric residual of the amplitude deficit after the kinetic margin has been 'used up.'

IV.3 The Three-Way Volatility Decomposition

The three volatilities can now be expressed as amplitude deficits with kinetic adjustments:

Derivation V.TVD · Three-Way Volatility Decomposition

Source volatility (amplitude deficit, no kinetic adjustment):

$$V_S = I_{min} = 1 - d_S = \Omega - d_S$$

(pure amplitude deficit, exact) (11)

The Source component never exceeds $d_S = 4/5$; the deficit $1/5$ is the full volatility

Boundary volatility (denominator deficit, no amplitude concept):

$$V_B = \sigma = 1 - d_B = \Omega - d_B$$

(pure denominator deficit, exact) (12)

The Boundary face is the axial governor; its deficit from Ω governs the Inverted Ratio

Remainder volatility (amplitude deficit minus kinetic margin):

$$V_R = \Lambda = (1 - d_R) - \Phi$$

(amplitude deficit reduced by kinetic margin) (13)

The Remainder component never exceeds d_R ; but the drive $\delta = d_R + \Phi$ slightly exceeds d_R

The Source and Boundary volatilities are pure geometric deficits — they equal exactly 1 minus their face's density. The Remainder volatility is a kinetically adjusted deficit — it equals 1 minus the face density, further reduced by the kinetic margin. This is the geometric consequence of the Remainder face's unique role: it is both the amplitude of the sine component (d_R) AND the structural threshold that the drive must exceed (d_R as the threshold). The Boundary face is purely a divisor (no amplitude role in the rotational plane). The Source face is purely an amplitude (no threshold role). The Remainder face is both amplitude and threshold simultaneously — which generates the kinetic adjustment to its volatility.

Part V — The Forbidden Ceiling as Exact Exhaust Rate

Part V derives the most structurally significant result of Essay VI: the permanent sub-lattice exhaust $\varepsilon_{snap} = 1/30$ per ordinal step carries information content exactly equal to $\log_2(3)$ bits — the Forbidden Ceiling of Essay III. This is not a numerical coincidence. It is a structural identity between the static informational bound (what T.NE forbids the static structure from achieving) and the kinetic exhaust rate (what the Inverted Ratio permanently expels per ordinal step). The kinetic structure exhausts the forbidden information at every step, which is precisely why it can never accumulate to the forbidden state.

V.1 The Exhaust Information Content

Derivation V.EIC · The Information Content of the Exhaust per Step

The sub-lattice exhaust per ordinal step is $\varepsilon_{snap} = 1/30$. As a fraction of the lattice grain δ :

$$\varepsilon_{snap} / \delta = (1/30) / (1/10) = 1/3 \quad (1)$$

The information content of this sub-lattice fraction — the number of bits required to specify this fraction within the unit lattice step — is:

$$\begin{aligned} I(\text{exhaust}) &= -\log_2(\varepsilon_{snap} / \delta) = -\log_2(1/3) = \\ \log_2(3) &= 1.5850 \text{ bits} \end{aligned} \quad (2)$$

Compare to the Forbidden Ceiling (Essay III, T.FC):

$$\text{Forbidden Ceiling} = \log_2(3) = 1.5850 \text{ bits} \quad (3)$$

$$I(\text{exhaust}) = \text{Forbidden Ceiling} \quad (\text{exact equality}) \quad (4)$$

V.2 T.FCE — The Forbidden Ceiling as Exhaust Rate

Theorem T.FCE · The Forbidden Ceiling as Exact Exhaust Rate

Per ordinal step of nothing's self-application, the sub-lattice structural exhaust carries information content equal to the Forbidden Ceiling:

$$-\log_2(\varepsilon_{\text{snap}} / \delta) = \log_2(3) = \text{Forbidden Ceiling} \quad (5)$$

Derivation:

$$\varepsilon_{\text{snap}} = G_{\text{raw}} - G_{\text{snap}} = 14/15 - 9/10 = 1/30 \quad (\text{Essay IV}) \quad (6)$$

$$\varepsilon_{\text{snap}} / \delta = (1/30) / (1/10) = 1/3 \quad (7)$$

$$-\log_2(1/3) = \log_2(3) = 1.5850 \text{ bits} = \text{Forbidden Ceiling} \quad (\text{T.FC, Essay III}) \quad (8)$$

Why is $\varepsilon_{\text{snap}}/\delta = 1/3$? Because $\varepsilon_{\text{snap}} = G_{\text{raw}} - G_{\text{snap}} = 14/15 - 9/10 = 28/30 - 27/30 = 1/30$, and $\delta = 1/10 = 3/30$. Therefore $\varepsilon_{\text{snap}} / \delta = (1/30)/(3/30) = 1/3$. The $1/3$ factor is exactly $1/d = 1/3$ where $d = 3$ is the Triad's dimension count (T.3D, Essay II).

V.3 The Structural Meaning of the Identity

The identity $I(\text{exhaust per step}) = \text{Forbidden Ceiling}$ has a precise structural meaning. The Forbidden Ceiling $\log_2(3)$ was derived in Essay III (T.FC) as the maximum face-label entropy achievable only in the degenerate uniform three-face partition — the state where each face occupies exactly $1/3$ of the structural space. This state is forbidden by T.NE (Essay I): the three faces must be strictly unequal. $\log_2(3)$ is the information content of the forbidden uniform state.

The kinetic exhaust rate being exactly $\log_2(3)$ bits per step means: at every ordinal step, the self-application of the Inverted Ratio permanently expels exactly the information content of the forbidden state. The exhaust IS the forbidden information, discharged one step at a time. This is the structural reason why nothing's kinetic structure can never accumulate toward the

forbidden state — because it continuously discharges the forbidden information rather than retaining it.

Principle P.FCE · The Structural Self-Consistency

If the kinetic structure retained rather than discharged the exhaust information:

$$\begin{aligned} \text{Retained } I(\text{exhaust}) \text{ per step} &= \log_2(3) \text{ bits} \\ \text{accumulated over } \tau \text{ steps} & \end{aligned} \quad (9)$$

After some ordinal count $\tau = \tau_{\text{forbidden}}$, the accumulated retained information would reach the Forbidden Ceiling $\log_2(3)$. At that moment, the static forbidden state would be informationally reproduced by the kinetic accumulation — T.NE would be violated.

The discharge of $\log_2(3)$ per step permanently prevents this accumulation. At every step, the forbidden information is expelled rather than retained. The kinetic exhaust is not a loss — it is a structural necessity. Nothing's self-application is calibrated to expel exactly the forbidden information per step because any retention of that information would violate T.NE and destroy the Triad.

The Forbidden Ceiling $\log_2(3)$ therefore plays two roles simultaneously: as the informational upper bound on the static structure (Essay III T.FC), and as the exact per-step exhaust rate of the kinetic structure (Essay VI T.FCE). These two roles are structurally identical — one read from the static register, the other from the kinetic register.

V.4 The Exhaust Accumulation per BRF and per Axial Closure

Derivation V.EAB · Exhaust Accumulation at Key Structural Periods

Per ordinal step:

$$I(\text{exhaust}) = \log_2(3) = 1.5850 \text{ bits} \quad (10)$$

Per Chronon ($\tau_0 = 10$ steps):

$$\begin{aligned} I(\text{exhaust}) &= 10 \times \log_2(3) = \log_2(3^{10}) = \\ \log_2(59049) &= 15.8496 \text{ bits} \end{aligned} \quad (11)$$

Per Boundary Resolution Frame ($R = 40$ steps):

$$I(\text{exhaust}) = 40 \times \log_2(3) = \log_2(3^{40}) = 63.40 \text{ bits} \quad (12)$$

Per full axial closure (100 steps):

$$\begin{aligned} I(\text{exhaust}) &= 100 \times \log_2(3) = \log_2(3^{100}) = \\ 158.50 \text{ bits} \end{aligned} \quad (13)$$

Structural reading: in one axial closure (100 ordinal steps), the kinetic structure expels $\log_2(3^{100})$ bits of forbidden information — 100 instances of the Forbidden Ceiling, one per step. The total exhaust over one complete axial period is $\log_2(3^{100})$ bits. Since 3^{100} is the information content of 100 independent degenerate-face assignments, the kinetic structure in one axial closure discharges the information equivalent of 100 separate forbidden states.

Part VI — The Saturation Threshold — Kinetic Derivation

Part VI derives the kinetic meaning of $N_{sat} = 25$. The value $N_{sat} = 1/P = 25$ was established in Essay II (V.25) as the static count of Structural Pixels tiling the unit Source x Boundary square. Essay VI now derives N_{sat} as the structurally forced saturation threshold of the helical worldline: the Pixel count at which the helix has exhausted the discrete address space of nothing's structural space and is compelled to reorganise. The derivation proceeds from the Pixel activation rate, through the axial closure period, to the structural event that forces coarse-graining.

VI.1 The Pixel Activation Rate

In one Chronon ($\tau_0 = 10$ ordinal steps), the helical worldline advances axially by exactly $\delta = 0.1$ (one lattice step) and visits $n = 10$ distinct Source positions — one per ordinal step within the Chronon. These $n = 10$ Source positions at one axial step constitute $n = 10$ distinct lattice cells in the Source x axial projection. A Structural Pixel $P = \sigma \times \delta$ spans $c_steps = \sigma/\delta = 4$ axial lattice steps at one Source column — it contains exactly $c_steps = 4$ lattice cells. The Pixel activation rate per Chronon:

Derivation V.PAR • The Pixel Activation Rate

Per Chronon ($\tau_0 = 10$ ordinal steps, axial advance = δ):

$$\begin{aligned} \text{Lattice cells covered per Chronon} &= n = 10 \\ &(\text{one per ordinal step}) \end{aligned} \tag{1}$$

$$\begin{aligned} \text{Lattice cells per Structural Pixel} &= c_steps = 4 \\ &(\sigma/\delta = 4 \text{ axial steps} \times 1 \text{ Source step}) \end{aligned} \tag{2}$$

$$\begin{aligned} \text{Pixel activation rate} &= n / c_steps = 10 / 4 = 5/2 \\ \text{pixels per Chronon} \end{aligned} \tag{3}$$

The activation rate $5/2$ is non-integer because $\gcd(n, c_steps) = \gcd(10, 4) = 2$, so $n/c_steps = 5/2$ is a reduced fraction. Integer pixel activation counts occur at Chronon multiples of $c_steps/\gcd(n, c_steps) = 4/2 = 2$:

$$\begin{array}{l} \text{After } 2 \text{ Chronons: } 2 \times (5/2) = 5 \text{ pixels} \\ \text{(first integer count)} \end{array} \quad (4)$$

$$\begin{array}{l} \text{After } 4 \text{ Chronons: } 4 \times (5/2) = 10 \text{ pixels} \\ (= n, = 40\% \text{ of } N_sat) \end{array} \quad (5)$$

$$\text{After } 8 \text{ Chronons: } 8 \times (5/2) = 20 \text{ pixels} \quad (6)$$

$$\text{After } 10 \text{ Chronons: } 10 \times (5/2) = 25 \text{ pixels} = N_sat \quad (7)$$

Full pixel saturation is reached at exactly $n = \tau_0 = 10$ Chronons.

VI.2 T.NSI — The Saturation-Closure Identity

The full $N_sat = 25$ pixel activation is reached at $n = \tau_0 = 10$ Chronons — the axial closure period of Essay V (V.HWP). This is not a coincidence: both quantities derive from n and c_steps alone, and their product is structurally forced.

Theorem T.NSI • The Saturation-Closure Identity

The full pixel saturation period equals the axial closure period:

$$\begin{array}{l} \text{Saturation period} = N_sat / \text{activation rate} = \\ 25 / (5/2) = 10 = \tau_0 \end{array} \quad (8)$$

Equivalently, rearranging:

$$\begin{array}{l} N_sat = \text{activation rate} \times \tau_0 = \\ (n / c_steps) \times n = n^2 / c_steps = \tau_0^2 / c_steps \end{array} \quad (9)$$

Proof of the identity $N_{sat} \times c_{steps} = n^2 = \tau_0^2$:

$$\begin{aligned} N_{sat} \times c_{steps} &= 25 \times 4 = 100 = 10^2 = \\ n^2 &= \tau_0^2 \quad \text{ok} \end{aligned} \tag{10}$$

Structural reading: $N_{sat} \times c_{steps}$ is the total lattice cells covered in one axial closure (n^2). The saturation threshold times the Pixel axial span equals the total address space of one axial closure. The Saturation Threshold, the Pixel geometry, and the Chronon are jointly determined: any one of them follows from the other two and n .

VI.3 T.NST — The Saturation Theorem

The saturation event is the structural event at Chronon $\tau_0 + 1 = 11$, at which the helix re-enters the axial level first visited at Chronon 1. This re-entry is the kinetic saturation event — the moment at which the Structural Pixel's discrete address capacity is structurally exhausted and sub-Pixel discrimination becomes informationally impossible.

Theorem T.NST • The Saturation Theorem

After $\tau_0 = n = 10$ Chronons, the helix has activated all $N_{sat} = 25$ Structural Pixels exactly once, covering $n^2 = 100$ distinct (Source, axial) lattice cells.

At Chronon $\tau_0 + 1 = 11$, the axial coordinate is:

$$z(11 \times \tau_0 \text{ ordinal steps}) = 11 \times \delta \equiv \delta \pmod{\Omega = 1} \tag{11}$$

This equals the axial coordinate at the start of Chronon 1. The rotational position at the start of every Chronon is $(d_S, 0)$ — since $\Omega \times \tau_0 = 2\pi$, the elliptic helix returns to the same rotational position after every Chronon (Essay V, V.HWO). Therefore:

$$\begin{aligned} \gamma(11 \times \tau_0) &= (d_S, 0, \delta) = \gamma(1 \times \tau_0) \\ &\text{(modulo axial closure)} \end{aligned} \tag{12}$$

Structural Pixel #1 — the pixel containing (Source position d_S , axial level δ) — has already been activated. The helix is about to register a SECOND activation of Pixel #1.

Why this forces coarse-graining:

The Structural Pixel $P = \sigma \times \delta = 1/25$ is the minimum addressable 2D cell, derived from the Minimum Bit-Budget (V.MB, Essay I; T.MBD, Essay III): any cell smaller than P would span fewer than 2 lattice positions in at least one direction, giving $\log_2(1) = 0$ bits of informational content. No *sub-P* address discrimination is structurally possible.

A second activation at Pixel #1 requires distinguishing the second visit from the first WITHIN Pixel #1 — a *sub-P* distinction. This has 0 bits of informational content by T.MBD. The second activation is informationally indistinguishable from the first within the Pixel's address resolution.

Second activation at Pixel #1 =>
sub-P discrimination required (13)

Sub-P discrimination => $\log_2(1) = 0$ bits
 (T.MBD, Essay III) (14)

0 bits => two activations indistinguishable =>
 coarse-graining forced (15)

The Saturation Threshold $N_{sat} = 25$ is the count of distinct Pixel activations before the first forced re-entry. At Chronon $\tau_0 + 1$, the $(N_{sat} + 1)$ -th activation re-enters Pixel #1, triggering structural coarse-graining. $N_{sat} = 25$ is therefore the maximum number of Structural Pixels the discrete lattice can address before being forced into a higher-resolution regime.

VI.4 The Three Saturation Constants and Their Identity

The saturation theorem rests on three structural constants, each derived from nothing's clearances, and their structural identity:

Derivation V.SCC · The Three Saturation Constants

From Essay I (T.UC, V.SP): the Boundary clearance step count

$$\begin{aligned} c_steps &= \sigma / \delta = 4 \\ (\text{Boundary clearance steps} &= \text{Pixel axial span}) \end{aligned} \quad (16)$$

From Essay V (T.TWR): the Chronon

$$\begin{aligned} \tau_0 &= n = 10 \\ (\text{Chronon} &= \text{lattice period, from } \text{ceil}(G_raw/\delta)) \end{aligned} \quad (17)$$

The Saturation Threshold:

$$\begin{aligned} N_sat &= \tau_0^2 / c_steps = n^2 / c_steps = \\ 100 / 4 &= 25 \end{aligned} \quad (18)$$

The mutual identity:

$$\begin{aligned} N_sat \times c_steps &= n^2 = \tau_0^2 \\ (\text{Saturation} \times \text{Pixel span} &= \text{Closure cells}) \end{aligned} \quad (19)$$

$c_steps = 4$ comes from $\sigma = 2/5$ (Boundary volatility, Essay VI Part I). $\tau_0 = 10$ comes from $\text{ceil}(G_raw/\delta)$ (Boundary inversion, Essay V Part II). $N_sat = 25$ comes from n^2/c_steps . All three are derived from $n = 10$ and $\sigma = 4 \times \delta = 2/5$ with zero free parameters. The coarse-graining threshold is structurally determined by nothing's own clearances.

Conclusion

What Has Been Derived

Essay VI has derived the geometry of nothing's kinetic volatility — the exact quantitative account of the permanent inaccessibility of each clearance fraction in the kinetic register — and has established three genuinely new structural results.

Part 0: The three kinetic volatilities $V_S = I_{min} = 1/5$, $V_B = \sigma = 2/5$, $V_R = \Lambda = 0.2984$ were defined as kinetic complements — the fractions of each axis permanently inaccessible to the helical worldline. The volatility hierarchy $V_B > V_R > V_S$ is the same as the clearance hierarchy from Essay I.

Part I: T.MVT (Minimum Volatility Theorem): $\sigma > I_{min}$ is the necessary and sufficient condition for $G_{raw} > d_R$ (positive kinetic output from the Inverted Ratio). This cross-clearance constraint is invisible in the static register and becomes visible only in the kinetic register. Foreclosure T.MVF: $\sigma = I_{min}$ is doubly foreclosed by T.MVT (zero output) and T.NE (equal face measures).

Part II: T.BRV: $R = \sigma/c = 40$ — the Boundary Resolution Frame is the ratio of the Boundary volatility to the causality speed. One *BRF* traverses exactly σ axially. In one *BRF*: 4 rotations, $\sigma = 0.4$ axial advance, $4/3$ units of sub-lattice exhaust, 63.4 *bits* of forbidden information discharged.

Part III: The exact arc coverage: $G_{raw}/\delta = 28/3$ per step; addressable 9 steps; sub-lattice gap $1/3$ per step. T.PSG: the gap is permanent because $3 = d$ (Triad dimension count) does not divide $28 = 4 \times 7$. Gap closure condition: $k = 3n$ (same as the Fundamental Stable Knot condition from Essay V). After 100 ordinal steps (full axial closure), the residual gap of $1/3$ persists permanently.

Part IV: T.VI (Volatility Identity): $\Lambda = (1 - d_R) - \Phi$. The Remainder volatility equals the Remainder amplitude deficit minus the kinetic margin. The three-way decomposition: Source and Boundary volatilities are pure geometric deficits $(1 - d_S, 1 - d_B)$; Remainder volatility is a kinetically adjusted deficit (the amplitude deficit reduced by Φ). The Remainder face's dual role as amplitude AND threshold generates this kinetic adjustment.

Part V: T.FCE (Forbidden Ceiling as Exhaust Rate): $-\log_2(\varepsilon_{\text{snap}}/\delta) = \log_2(3) = \text{Forbidden Ceiling}$. Per ordinal step, nothing's self-application expels exactly the forbidden information. The static informational bound and the kinetic exhaust rate are the same structural constant in two registers. The kinetic exhaust prevents accumulation toward the forbidden state by discharging it continuously.

Part VI: T.NSI (Saturation-Closure Identity): $N_{\text{sat}} = \tau_0^2/c_{\text{steps}} = n^2/c_{\text{steps}} = 25$. Pixel activation rate $= n/c_{\text{steps}} = 5/2$ per Chronon (V.PAR). All 25 Structural Pixels activated in exactly $\tau_0 = 10$ Chronons (one full axial closure). T.NST: at Chronon $\tau_0 + 1 = 11$, the helix re-enters Pixel #1 at axial level $\delta \pmod{1}$. Sub-Pixel discrimination requires $\log_2(1) = 0 \text{ bits}$ (T.MBD, Essay III) — informationally impossible. Coarse-graining is structurally forced at the N_{sat} -th Pixel activation.

The Volatility Terminal Architecture

Theorem T.VTA • The Volatility Terminal Architecture — Essay VI

Nothing's kinetic volatility structure, completely derived from the three clearances:

$$\begin{aligned} \text{T.MVT: } \sigma &> I_{\text{min}} \\ (\text{cross-clearance constraint for } G_{\text{raw}} > d_R) \end{aligned} \tag{18}$$

$$\begin{aligned} \text{T.BRV: } R &= \sigma/c = 40 \\ (\text{BRF} = \text{Boundary volatility} / \text{causality speed}) \end{aligned} \tag{19}$$

$$\begin{aligned} \text{T.PSG: } \text{gap} &= 1/3 \text{ per step} \\ (\text{sub-lattice gap from } \gcd(28, 3) = 1) \end{aligned} \tag{20}$$

$$\text{T.VI: } \Lambda = (1-d_R) - \Phi$$

(Volatility Identity) (21)

$$\text{T.FCE: } I(\text{exhaust/step}) = \log_2(3) = \text{Forbidden Ceiling}$$

(exact exhaust rate) (22)

$$\text{V.NSC: } N_{\text{sat}} = n^2/c_{\text{steps}} = 25$$

(saturation threshold from lattice) (23)

All six results are derived from the three clearances $I_{\min} = 1/5$, $\sigma = 2/5$, $\Lambda = 0.2984$ and the lattice constants $n = 10$, $d = 3$. Zero free parameters. The geometry of volatility is the kinetic register's reading of what the static register derived as structural margins. The two registers are the same architecture seen from opposite sides of the ordinal sequence.

Bibliography

The following works are cited for structural isomorphism confirmation only. None serve as derivational foundations for the present framework. All results herein are derived from the internal logic of the Triad and the normalised field alone.

Internal series references

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ADDENDUM: UNIVERSAL TERMS OF ENGAGEMENT FOR THE RESIDUAL SUITE

Preamble This addendum serves as the binding and immutable interpretive constraint specifically for the **Residual Suite** (Validation Protocols and Ledger Audits for the 20% Minimum Bit-Budget, 30% Ontological Gap, and 40% Structural Volatility). Its purpose is to formalize the absolute universality of these geometric clearances, explicitly foreclosing any attempt to restrict them to a specific domain, timeframe, or material.

CONSTRAINT 1: Phase Invariance (The Permanent Mechanics)

The three residuals ($I_{min} \approx 0.2, \Lambda \approx 0.298, \sigma = 0.4$) are not pre-kinetic artifacts or temporary conditions of the primordial stasis. They are the permanent, load-bearing mechanical clearances of the universe. Analysts are strictly prohibited from treating these gaps as confined to Phase I. They operate continuously across all phases—preventing the Deterministic Crystal in Phase II and acting as the Velocity/Saturation blocks that mandate perpetual novelty in Phase III. The gaps are the permanent operating tolerance of the engine.

CONSTRAINT 2: Scale Invariance (The Scalar Mandate)

The structural clearances apply with absolute isomorphic fidelity at every scalar resolution of the **Veldt**. Readers and analysts are prohibited from treating the residuals as exclusively "quantum" phenomena or exclusively "cosmological" phenomena. The 40% Structural Volatility is identically the source of Heisenberg's Uncertainty Principle at the micro-scale, the mandated tolerance in macroscopic precision engineering, and the limit of resolution in computational mathematics. The math is scale-invariant.

CONSTRAINT 3: Substrate Neutrality (The Purge of Material Bias)

The residuals are pure mathematical inevitabilities of a discrete lattice. They must never be reified by attributing them to a specific material substrate. The 20%, 30%, and 40% clearances govern biological ecosystems, artificial intelligence networks, silicon processing, and the astrophysical void equally. Importing material, biological, or anthropic properties to explain the existence of these gaps is a prohibited circularity.

CONSTRAINT 4: Epistemic Foreclosure (Anti-Reification of Uncertainty)

Regardless of the phase, scale, or substrate being analyzed, no reader may interpret these three residuals as measurement errors, statistical noise, or "epistemic limits" arising from observer ignorance. They are derived **Mechanical Necessities**. To claim these gaps can be "closed" by better measurement, or to treat them as probabilistic noise, is to fundamentally misunderstand the engine. **The residuals are not what the engine lacks — they are what the engine is.**

CONSTRAINT 5: Final Demarcation and Logical Necrosis

These Terms of Engagement are non-negotiable. Any attempt to localize these clearances to a specific phase, restrict them to a specific scale, tie them to a specific substrate, or dismiss them as probabilistic contingencies constitutes a formal epistemic breach classified as **Logical Necrosis**.